

Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Python · SQL · Power BI



Dataset at a Glance

3,900

Transactions

18

Features

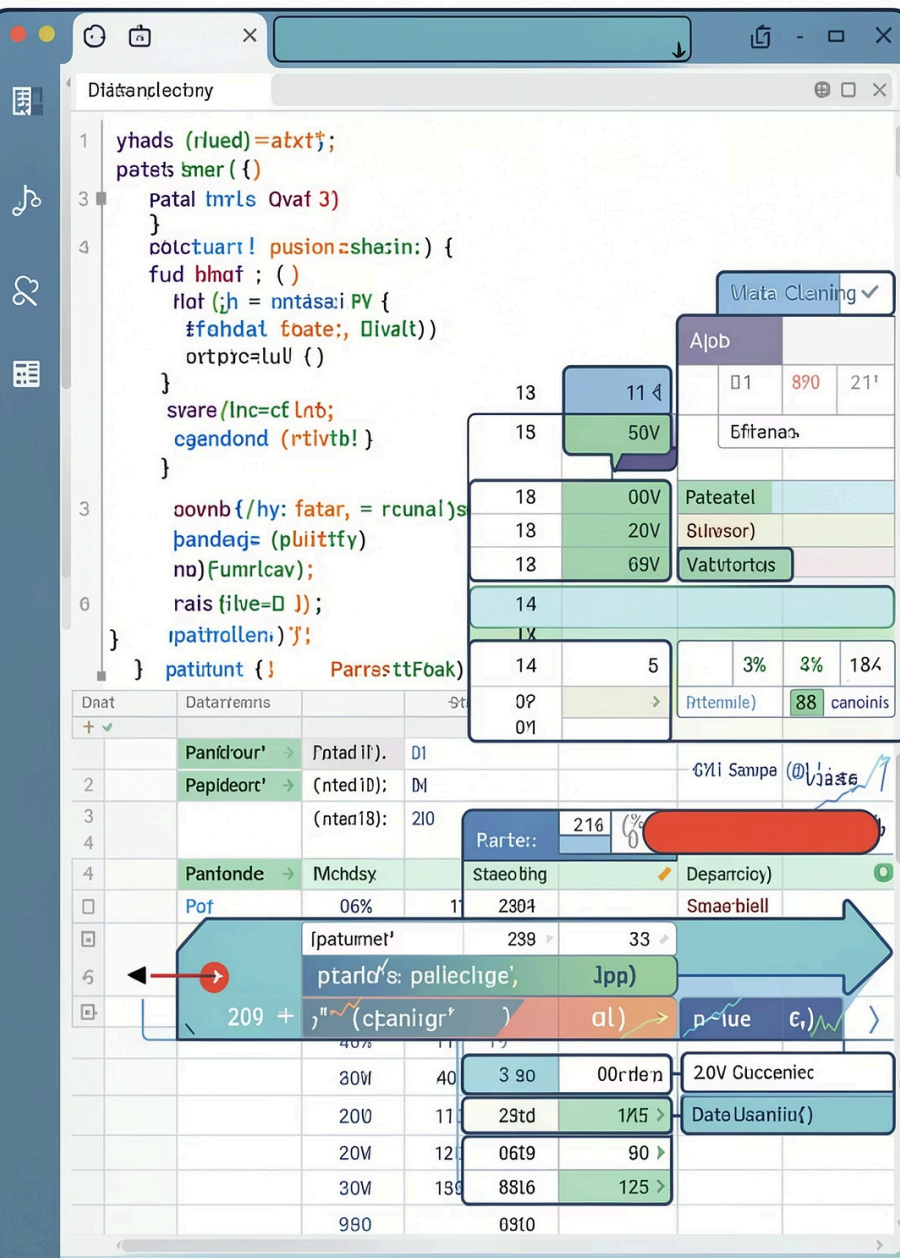
37

Missing Values

Review Rating only

\$59.76

Avg. Purchase



Data Preparation in Python

→ Load & Explore

pandas, df.info(), describe()

→ Clean

Median imputation for Review Rating

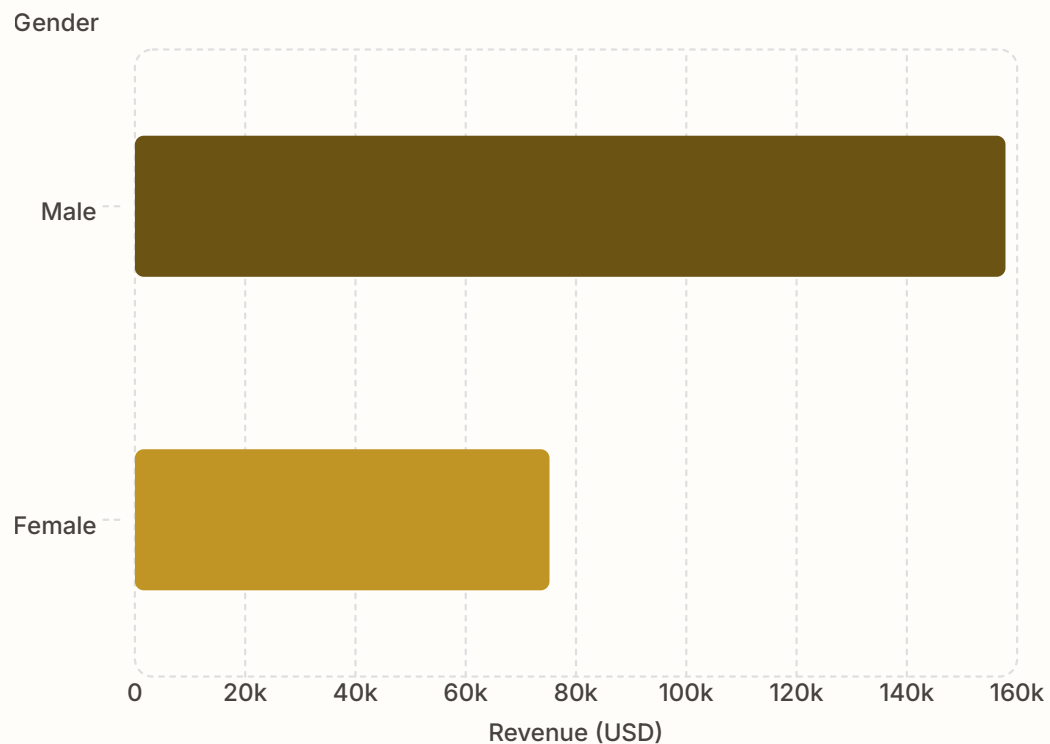
→ Engineer

age_group, purchase_frequency_days

→ Load to DB

PostgreSQL for SQL analysis

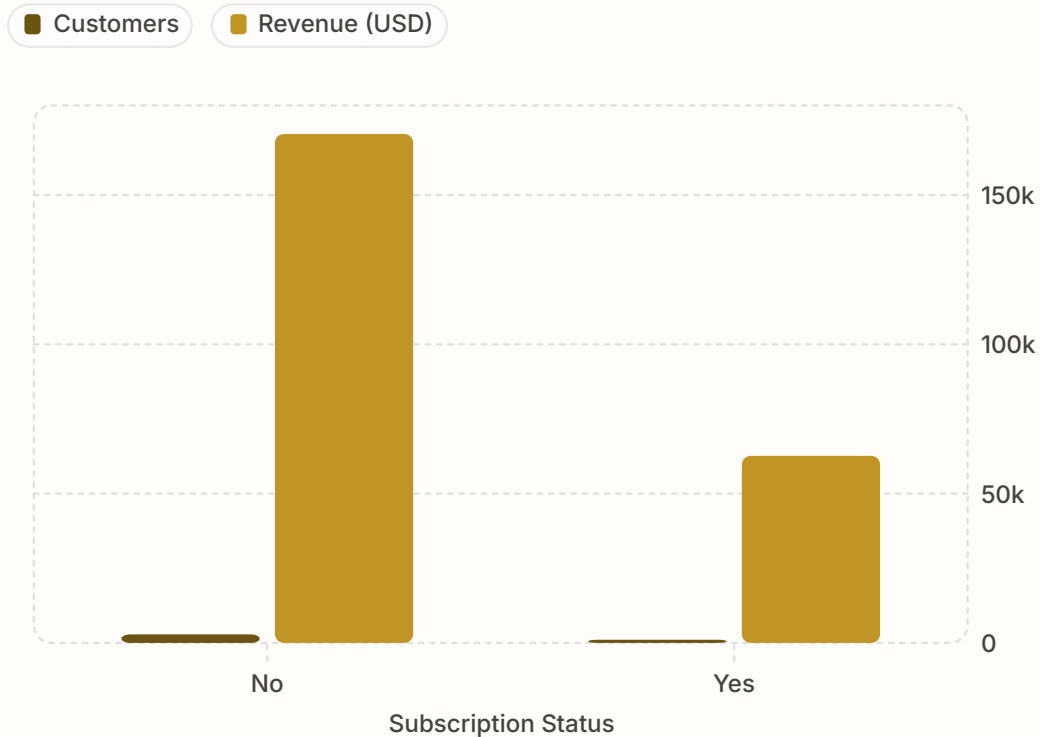
Revenue by Gender



Key Insight

Male customers generated **2.1× more revenue** than female customers — \$157,890 vs. \$75,191.

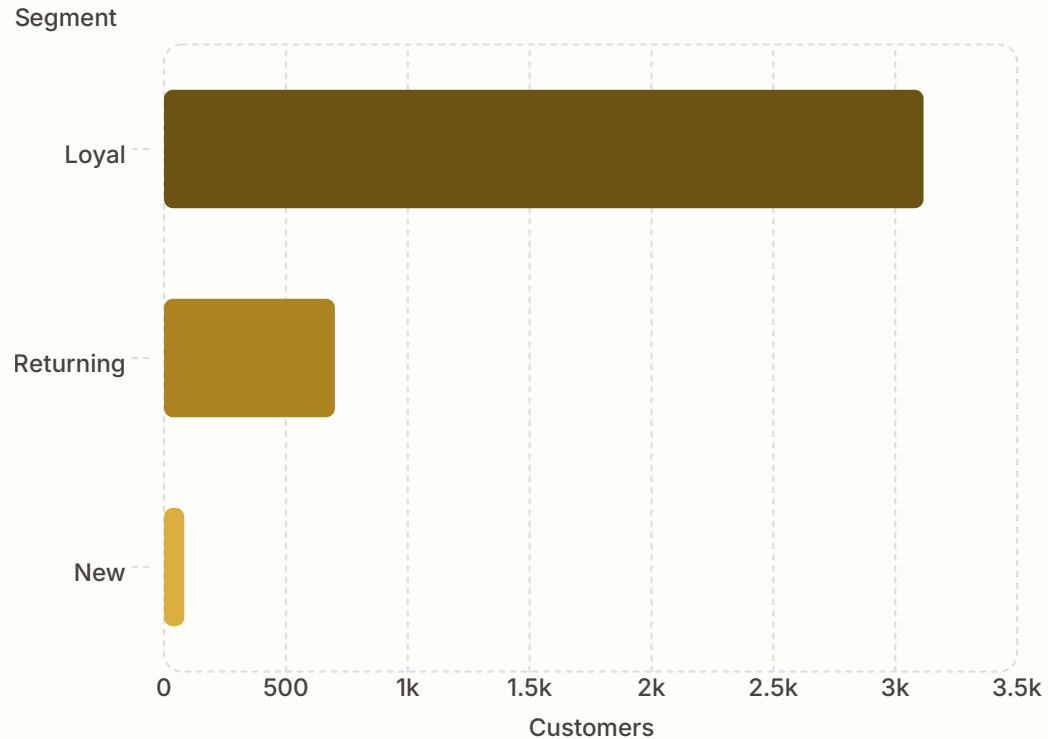
Subscribers vs. Non-Subscribers



Key Insight

Non-subscribers are **73%** of customers and drive **\$170K revenue**. Subscribers spend nearly the same avg (\$59.49 vs \$59.87).

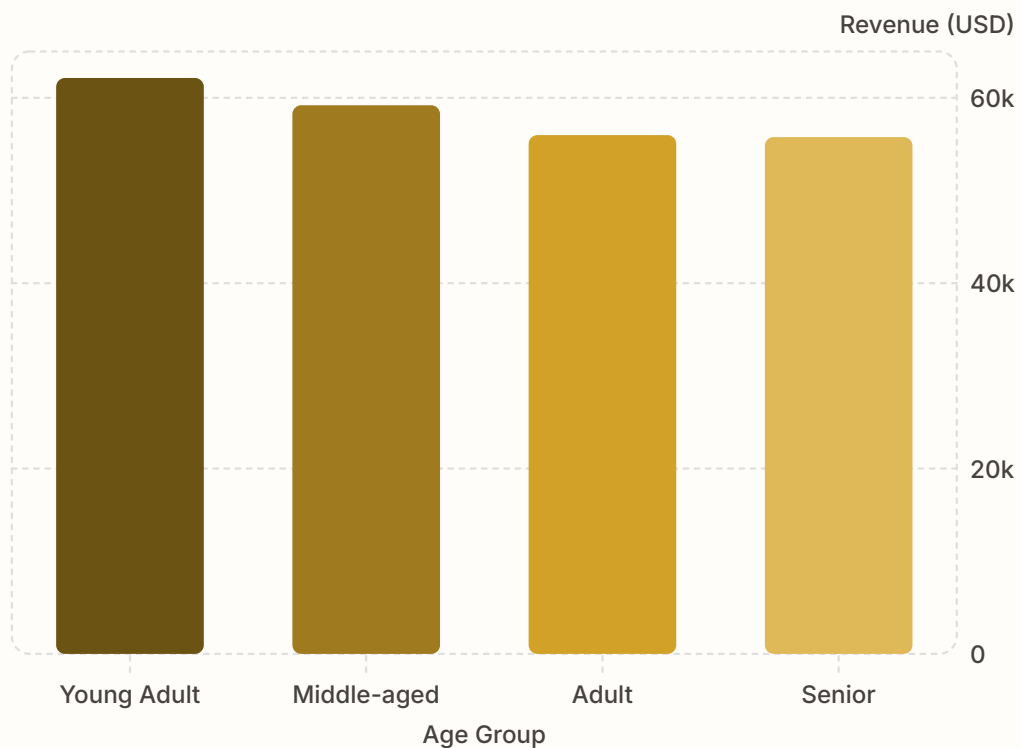
Customer Segmentation



Key Insight

80% of customers are Loyal (3,116). Only 83 new customers — opportunity to grow the base.

Revenue by Age Group



Key Insight

Young Adults lead revenue at **\$62,143**. Revenue is fairly balanced across all age groups.

Top Products by Category



Accessories

Jewelry · Sunglasses · Belt



Clothing

Blouse · Pants · Shirt



Footwear

Sandals · Shoes · Sneakers



Outerwear

Jacket · Coat

Power BI Dashboard



\$3.9K

Customers

\$59.76

Avg Purchase

3.75

Avg Rating

27%

Subscribers

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers → Loyal segment

Review Discounts

Balance sales lift with margin control

Targeted Marketing

Focus on Young Adults & express-shipping users